

2020 AMC 10A Problem 25 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10A Problem 25. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10A Problem 25

Problem:

Jason rolls three fair standard six-sided dice. Then he looks at the rolls and chooses a subset of the dice (possibly empty, possibly all three dice) to reroll. After rerolling, he wins if and only if the sum of the numbers face up on the three dice is exactly 7. Jason always plays to optimize his chances of winning. What is the probability that he chooses to reroll exactly two of the dice?

Answer Choices:

A. $\frac{7}{36}$

B. $\frac{5}{24}$

C. $\frac{2}{9}$

D. $\frac{17}{72}$

E. $\frac{1}{4}$

Solution:

Consider the probability that rolling two dice gives a sum of s , where $s \leq 7$. There are $s - 1$ pairs that satisfy this, namely $(1, s - 1), (2, s - 2), \dots, (s - 1, 1)$, out of $6^2 = 36$ possible pairs. The probability is $\frac{s - 1}{36}$.

Therefore, if one die has a value of a and Jason rerolls the other two dice, then the probability of winning is $\frac{7-a-1}{36} = \frac{6-a}{36}$.

In order to maximize the probability of winning, a must be minimized. This means that if Jason rerolls two dice, he must choose the two dice with the maximum values.

Thus, we can let $a \leq b \leq c$ be the values of the three dice. Consider the case when $a + b < 7$. If $a + b + c = 7$, then we do not need to reroll any dice. Otherwise, if we reroll one die, we can reroll any one die in the hope that we get the value that makes the sum of the three dice 7. This happens with probability $\frac{1}{6}$. If we reroll two dice, we will roll the two maximum dice, and the probability of winning is $\frac{6-a}{36}$, as stated above.

However, $\frac{1}{6} > \frac{6-a}{36}$, so rolling one die is always better than rolling two dice if $a + b < 7$.

Now consider the case where $a + b \geq 7$. Rerolling one die will not help us win since the sum of the three dice will always be greater than 7. If we reroll two dice, the probability of winning is, once again, $\frac{6-a}{36}$. To find the probability of winning if we reroll all three dice, we can let each die have 1 dot and find the number of ways to distribute the remaining 4 dots. By stars and bars, there are $\binom{6}{2} = 15$ ways to do this, making the probability of winning $\frac{15}{6^3} = \frac{5}{72}$.

In order for rolling two dice to be more favorable than rolling three dice, $\frac{6-a}{36} > \frac{5}{72} \rightarrow a \leq 3$. Thus, rerolling two dice is optimal if and only if $a \leq 3$ and $a + b \geq 7$. The possible triplets (a, b, c) that satisfy these conditions, and the number of ways they can be permuted, are:

$(3, 4, 4) \rightarrow 3$ ways

$(3, 4, 5) \rightarrow 6$ ways

$(3, 4, 6) \rightarrow 6$ ways

$(3, 5, 5) \rightarrow 3$ ways

$(3, 5, 6) \rightarrow 6$ ways

$(3, 6, 6) \rightarrow 3$ ways

$(2, 5, 5) \rightarrow 3$ ways

$(2, 5, 6) \rightarrow 6$ ways

$(2, 6, 6) \rightarrow 3$ ways

$(1, 6, 6) \rightarrow 3$ ways

There are $3 + 6 + 6 + 3 + 6 + 3 + 3 + 6 + 3 + 3 = 42$ ways in which rerolling two dice is optimal, out of $6^3 = 216$ possibilities, Therefore, the probability that Jason will reroll two dice is:

$$\frac{42}{216} = \boxed{(A) \frac{7}{36}}$$

OR

We conclude all of the following after the initial roll:

1. Jason rerolls exactly zero dice if and only if the sum of the three dice is 7, in which the probability of winning is always 1.
2. If Jason rerolls exactly one die, then the sum of the two other dice must be 2, 3, 4, 5, or 6. The probability of winning is always $\frac{1}{6}$, as exactly 1 of the 6 possible outcomes of the die rerolled results in a win.
3. If Jason rerolls exactly two dice, then the outcome of the remaining die must be 1, 2, 3, 4, or 5. Applying casework to the remaining die produces the following table:

Remaining Die	Sum Needed for the Two Dice Rerolled	Probability of Winning
1	6	$\frac{5}{36}$
2	5	$\frac{4}{36}$
3	4	$\frac{3}{36}$
4	3	$\frac{2}{36}$
5	2	$\frac{1}{36}$

The probability of winning is at most $\frac{5}{36}$.

4. If Jason (re)rolls all three dice, then the probability of winning is always $\frac{\binom{6}{2}}{6^3} = \frac{15}{216} = \frac{5}{72}$. For the denominator, rolling three dice gives a total of $6^3 = 216$ possible outcomes. For the numerator, this is the same as counting the ordered triples of positive integers (a, b, c) for which $a + b + c = 7$. Suppose that 7 balls are lined up in a row. There are 6 gaps between the balls, and placing dividers in 2 of the gaps separates the balls into 3 piles. From

left to right, the numbers of balls in the piles correspond to a , b , and c , respectively. There are $\binom{6}{2} = 15$ ways to place the dividers. Note that the dividers' positions and the ordered triples have one-to-one correspondence, and $1 \leq a, b, c \leq 6$ holds for all such ordered triples.

The optimal strategy is that:

- If Jason needs to reroll at least zero dice to win, then he rerolls exactly zero dice.
- If Jason needs to reroll at least one die to win, then he rerolls exactly one die.
- If Jason needs to reroll at least two dice to win, then he rerolls exactly two dice if and only if the probability of winning is greater than $\frac{5}{72}$, the probability of winning for rerolling all three dice.,

The first three cases in the table above satisfy this requirement. We will analyze these cases by considering the initial outcomes of the two dice rerolled. Note that any of the three dice can be the remaining die, so we need a factor of 3 for all counts in the third column:

Remaining Die	Initial Outcomes of the Two Dice Rerolled	# of Ways
1	Both are in $\{6\}$, not necessarily distinct.	$3 \cdot 1^2 = 3$
2	Both are in $\{5, 6\}$, not necessarily distinct.	$3 \cdot 2^2 = 12$
3	Both are in $\{4, 5, 6\}$, not necessarily distinct.	$3 \cdot 3^2 = 27$

Finally, the requested probability is

$$\frac{3 + 12 + 27}{6^3} = \frac{42}{216} = \boxed{(A) \frac{7}{36}}$$

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